

11. A group of students investigate how the surface area of a falling paper cake case affects its terminal speed.

- Cake case 1 has a mass of 0.5 g and a surface area of 100 cm².
- Cake case 1 is dropped from a height of 1.80 m but only timed over the final 1.50 m of the fall.

The students' results are shown in the table below.

Drop time (s)			Mean drop time (s)	Drop distance (m)
Attempt 1	Attempt 2	Attempt 3		
0.96	0.92	0.94		1.50

(a) (i) The students decide there are no anomalies. Explain why. [1]

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(ii) **Complete the table** to show the mean drop time. [1]
Space for calculation.

- (b) The experiment is repeated with cake case 2.
It has the same shape and the same mass as cake case 1.
However, cake case 2 has a surface area of 50 cm².
The students correctly calculate the terminal speed for both cake cases.

Cake case 1		
Mass (g)	Surface area (cm ²)	Terminal speed (m/s)
0.5	100	1.6

Cake case 2		
Mass (g)	Surface area (cm ²)	Terminal speed (m/s)
0.5	50	2.3



- (i) A cake case reaches terminal speed when its weight is balanced by air resistance. **Tick (✓) the three correct statements.** [3]

- Cake case 2 has the same terminal speed as cake case 1. ☐
- Cake cases 1 and 2 have identical weight. ☐
- At terminal speed, cake case 1 experiences a greater value of air resistance than cake case 2. ☐
- At terminal speed, both cake cases experience identical values of air resistance. ☐
- At terminal speed, cake case 1 experiences a smaller value of air resistance than cake case 2. ☐
- At terminal speed, both cake cases have zero acceleration. ☐

- (ii) Before the experiment was carried out the students made the following prediction:
“If the surface area of the cake case is halved its terminal speed will double.”
Use data from the tables on the previous page to explain whether their prediction was correct. [2]

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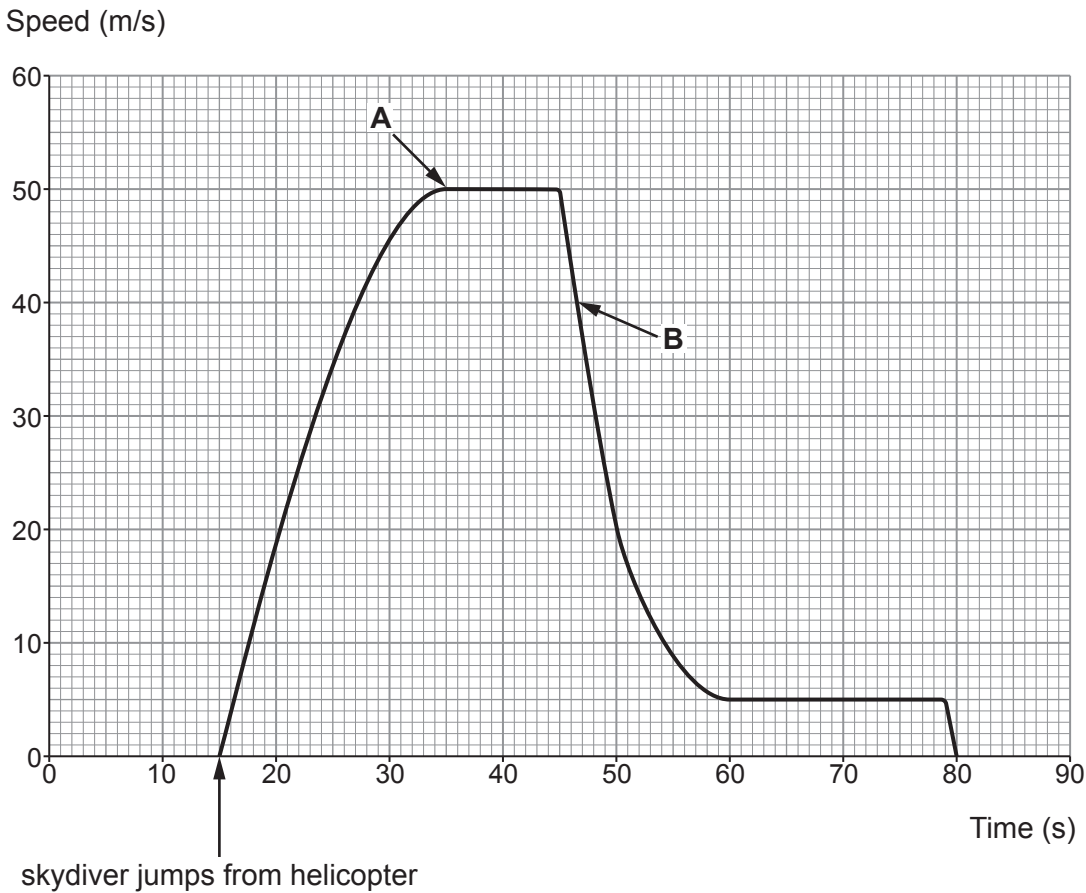
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- (c) A skydiver sits at the doorway of a helicopter for 15 s before jumping. His speed is recorded and displayed on the graph below.



Tick (✓) the **three** correct statements.

[3]

- The skydiver lands on the ground 80 s after jumping from the helicopter. ☐
- The terminal speed after the parachute is opened is $\frac{1}{10}$ th of the terminal speed before the parachute is opened. ☐
- The skydiver's weight is greatest at the point labelled **B** on the graph. ☐
- The parachute is opened 30 s after the skydiver leaves the helicopter. ☐
- At point **B** the weight of the skydiver is greater than the air resistance. ☐
- At point **A** the skydiver stops accelerating. ☐

END OF PAPER

